Analyzing Movement Dynamics in Elite Female Soccer Athletes

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Introduction

Motivation: As women's soccer grows, understanding the physiological demands on elite female athletes is crucial to enhancing performance and minimizing injury risks. However, research specific to female athletes has historically been limited.

Goal: Quantify movement patterns in elite female soccer athletes and examine links to performance and health outcomes.

Approach:
- Construct a 3D "quantile cube" based on velocity, acceleration, and angle.
- Use Hellinger Distance metric to detect significant differences in distributions.
- Use Principal Component Analysis (PCA) to identify key movement components and anomalies.
- Apply Dirichlet-Multinomial Regression to assess the effects of time, position, and soreness movements.

Methods

Data: GPS data from 9 elite female soccer athletes across two halves of 22 matches. All athletes logged over 25 minutes per half in at least 5 matches. Fit For 90 daily surveys for each player throughout the season.

GPS Data: One data point per second capturing the longitude and latitude coordinates of each player's location during play.

Data Transformation:
- GPS coordinates were transformed into (x, y) field coordinates (in meters) relative to the center.
- Velocity and acceleration were derived using a third-degree spline and calculated via first and second derivatives.

Movement Metrics:
- Velocities < 0.01 m/s and accelerations < 0.001 m/s² were set to 0.
- Velocity and acceleration were log-transformed to correct skewness.
- Movement angle was calculated from the difference between velocity and acceleration angles, adjusted to a range of -180° to 180°.

Methods

Data Object: **Quantile Cube**
- We established five quantiles for velocity and acceleration and four for movement angle (-35° baseline, representing forward, right, backward, and left movement) from the full dataset.
- From these boundaries, we construct a quantile cube for each half of each game for each player with dimensions for velocity, acceleration, and angle.
- The quantile cube can be viewed in terms of counts, representing the number of seconds (divided by 10) spent within each quantile, or proportions, representing the proportion of time spent in a section of the cube.

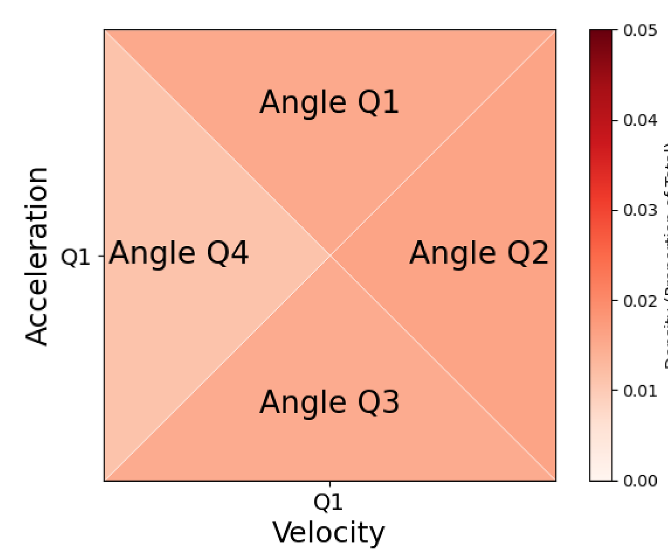

Fig 1. Example Piece of Quantile Cube

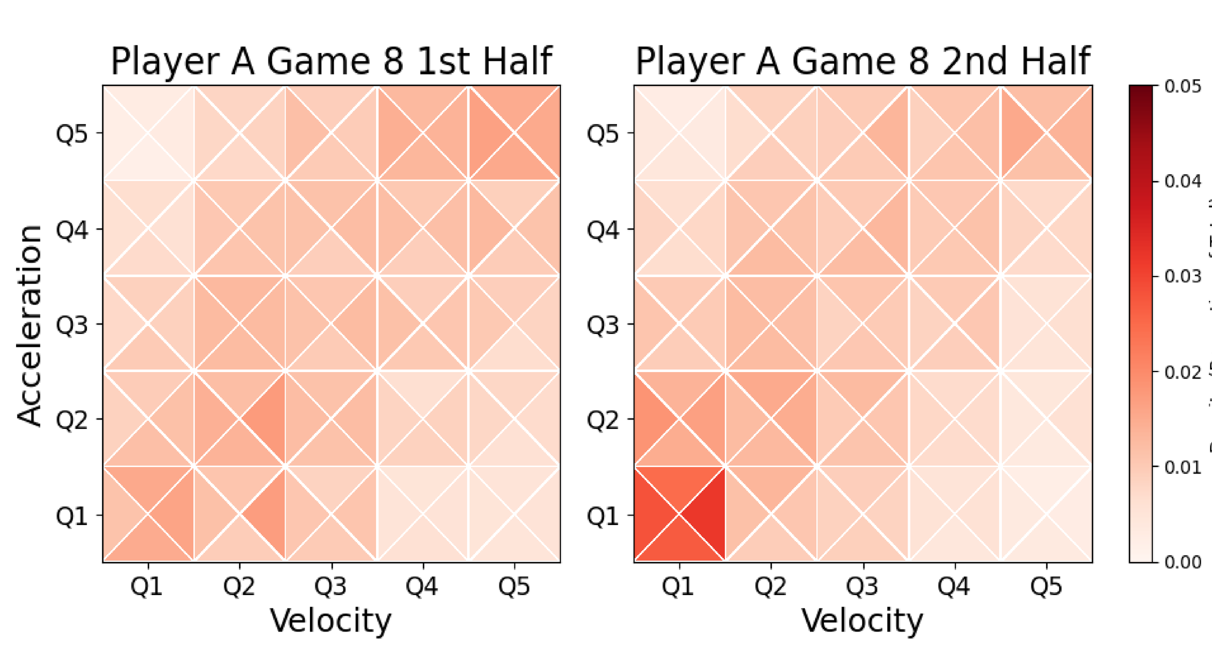

Fig 2. Quantile Cubes for Player A Game 8 1st and 2nd Halves

Dirichlet-Multinomial Distribution
- In our data matrix $\boldsymbol{Y}$, we have $n = 294$ quantile cubes (rows), each flattened into $d = 100$ dimensions (columns).
- To account for overdispersion, we assume each observation:

$$\boldsymbol{y}_i \sim Multinomial(N_i, \boldsymbol{\pi})$$
$$\boldsymbol{\pi} = (\pi_1, \ldots, \pi_d) \sim Dirichlet(\boldsymbol{\alpha})$$

where $N_i = \sum_{j=1}^{d} y_{ij}$ is the total count for observation i, $\pi_j > 0$ for all $j \in \{1, \ldots, d\}$, $\sum_{j=1}^{d} \pi_j = 1$, and $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_d)$ are positive concentration parameters.

Results

First vs. Second Half: Using the Hellinger distance metric, there is a statistically significant difference in the distribution of movements between the first and second half for each game for each player.

PCA: We reduce the data into 5 components explaining 71.4% of variance. PC3 specifically identifies a game that resulted in a season-ending loss, exhibiting higher proportions of time spent at low velocities with high acceleration.

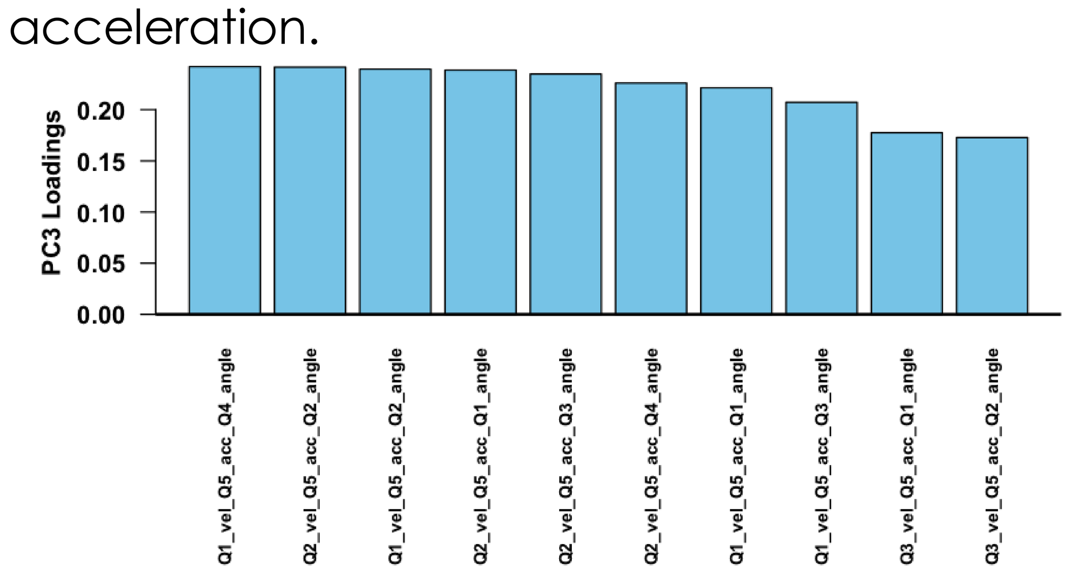

Fig 3. Top 10 PC3 Loadings

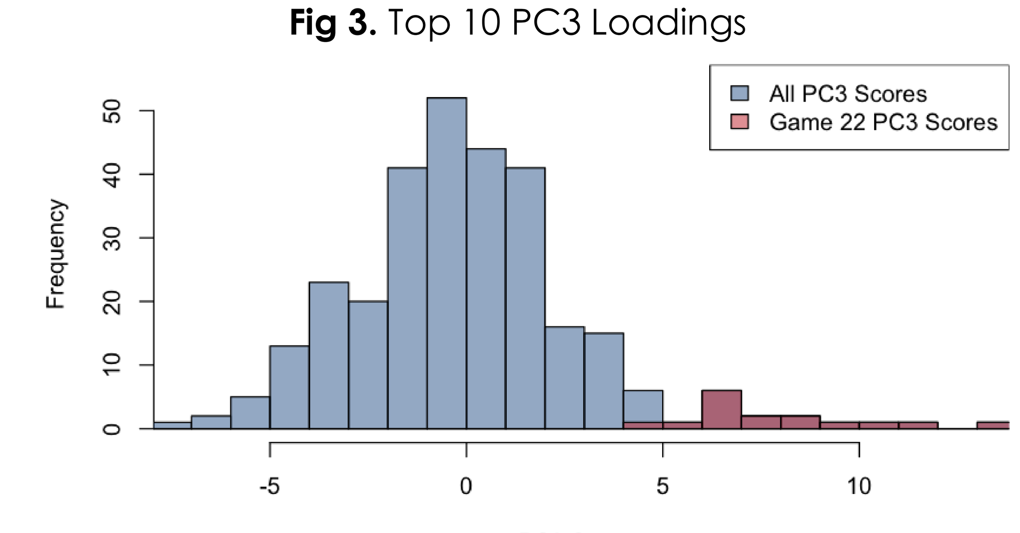

Fig 4. Histogram of PC3 Scores

Dirichlet-Multinomial Regression Model: Let $\boldsymbol{X} = (x_{ik})_{n \times p}$ be the design matrix. We assume parameters $\boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_d)$ in the Dirichlet-Multinomial model depend on the covariates with $\alpha_j(\boldsymbol{y}_i) = \exp(\beta_{j0} + \sum_{k=1}^{p} \beta_{jk} x_{ik})$ where β_{j0} is a constant for each j and β_{jk} is the coefficient for the j^{th} dimension with respect to the k^{th} covariate.

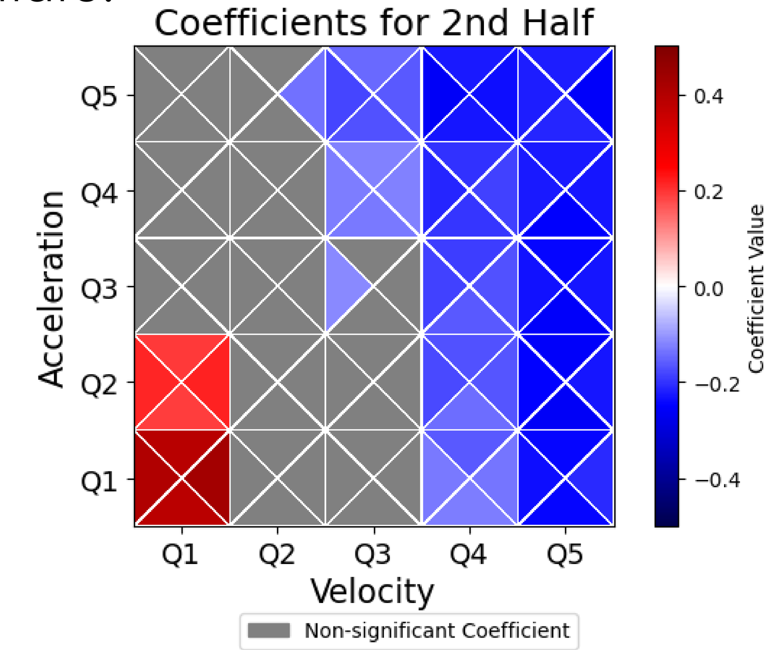

Fig 5. DMM Coefficients for 2nd Half (Baseline Category: 1st Half)

Results

DMM Model: half (1st and 2nd), position group (defender, midfielder, and forward), soreness rating 2 days before match day, MD-2 (Likert scale -3 to 3).

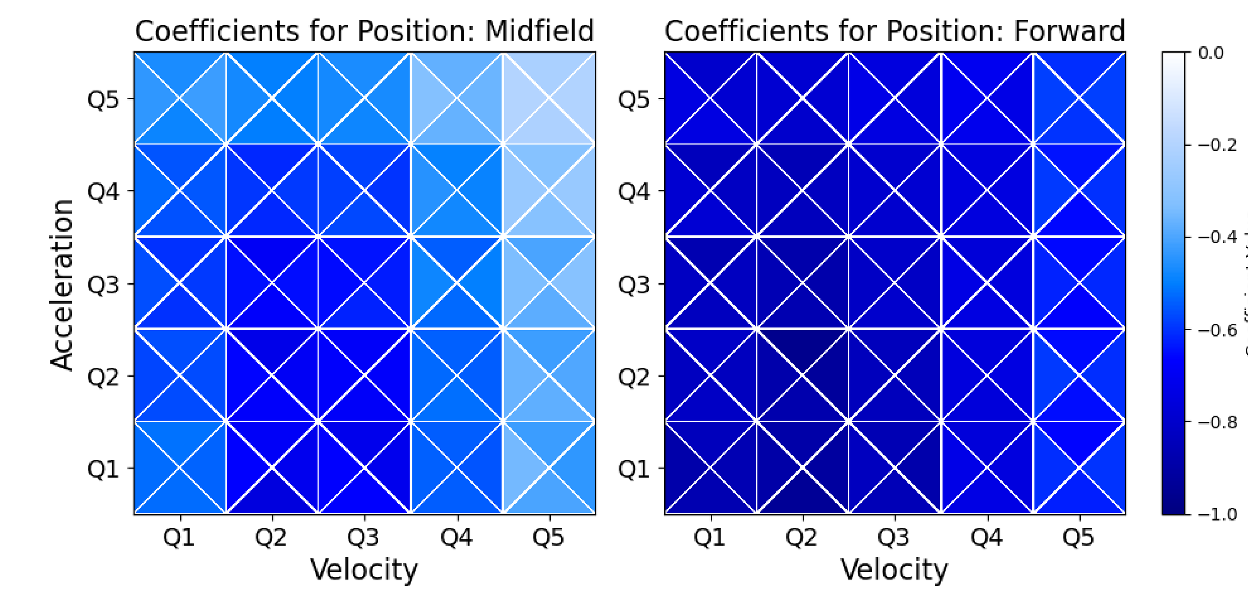

Fig 6. DMM Coefficients for Position Groups (Baseline Category: Defender) All coefficients are significant.

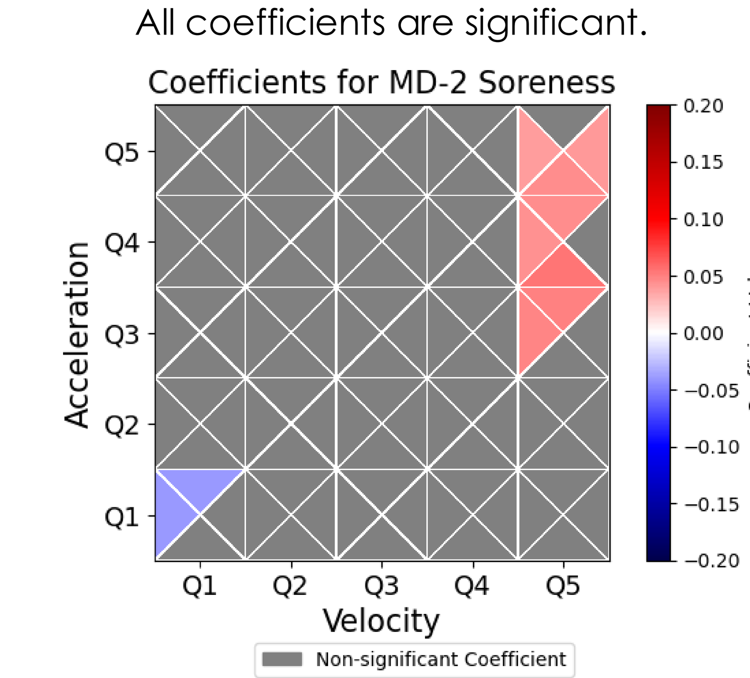

Fig 7. DMM Coefficients for MD-2 Soreness

Conclusions & Future Directions

DMM applied to the quantile cubes accurately models the impact of time, position, and soreness:
- In the 2nd half, players spend more time at low velocities and accelerations and less time at higher velocities and accelerations compared to the 1st half.
- Forwards and midfielders play less time in general than defenders.
- Players who are less sore (+3) MD-2 spend more time at higher velocities and acceleration.

Future: Extend the DMM using fiducial inference to derive parameter estimates without relying on prior distributions.

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